

Aryan Ahlawat

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Queen's University, Kingston, ON

Bachelor of Computing (Honours) — AI specialization stream — Economics Minor

2024 – 2028

Experience:

Software Developer Co-op — Co-operators — AWS Lambda, Lex, Python, Postman

May 2026 – Aug 2026

- Currently working on developing **RESTful** Prefill API integrations to enable real-time auto-population of insurance quoting forms, orchestrating data retrieval across multiple internal services and third-party endpoints to eliminate manual entry and reduce quoting workflow completion time

Design Team Engineer — QMIND — Langchain, HuggingFace, RAGAS, Python

Sept 2025 – April 2026

- Developing a Cognitive **Retrieval-Augmented Generation (RAG)** system, utilizing **LangChain** agents to enable dynamic reasoning and multi-hop question answering for client data
- Engineered a hybrid retrieval pipeline combining BM25 **sparse search** and **bi-encoder** dense embeddings, optimizing context recall for multi-hop question answering tasks
- Integrated **reciprocal rank fusion (RRF)** and a **cross-encoder** re-ranking model from Hugging Face to enhance retrieval precision and accuracy

Machine Learning Co-op — Acetech — PyTorch, Pandas, SQL, NumPy

May 2025 – Aug 2025

- Developed Multi-Layer Perceptron (**MLP**) regression models using **PyTorch** to predict lab test durations, optimizing hyperparameters to achieve a 12% reduction in Mean Squared Error Loss compared to the previous baseline
- Preprocessed and engineered features from an extensive SQL dataset; engineered automated data cleaning pipelines using **Pandas**, **SQLAlchemy**, and **NumPy** to impute missing values and detect outliers

Drive Control Developer — Queen's Knights Robotics — FreeRTOS, MATLAB, C

Oct 2024 – March 2025

- Programmed low-level robotic **drive-control systems** in C, implementing **PID controllers** to ensure smooth acceleration and precise velocity control
- Developed **motion algorithms** that improved autonomous navigation efficiency, reducing path deviations for competition tasks by almost 20%

Research Assistant — University of Toronto — COBWEB

Jun 2023 – Feb 2024

- Developed **agent-based simulations** using the **COBWEB** framework to model and analyze complex economic decision-making scenarios utilizing club theory and simulating cooperative vs. competitive behavior
- Enhanced simulation accuracy by refining agent parameters and contributed comprehensive technical documentation for the research group

Projects:

Churn Classification Engine — Python, Scikit-Learn, Seaborn

Nov 2025

- Developed an end-to-end classification pipeline incorporating Logistic Regression and Random Forest ensembles successfully segmenting 7,000+ users by probability of churn with a 0.82 F1 score.

VisualizeIt — (Winner of Mayor's Innovation Award) — YOLOv8, PyTorch, Stable Diffusion

Feb 2025

- Engineered a real-time computer vision pipeline utilizing YOLOv8 (Nano) for millisecond-latency object detection, serving as the initialization anchor for robust OpenCV CSRT tracking
- Integrated Stable Diffusion inpainting to generate context-aware texture overlays, dynamically mapping stylized graphics onto detected regions with high geometric consistency.

Languages: Python, Java, C, SQL, Flutter/Dart, JavaScript, PyTorch, Langchain, Scikit-learn

Certifications: Oracle Cloud Infrastructure Foundations, Microsoft AZ-900, NVIDIA Building RAG Agents with LLMs

Technologies: GitHub, Microsoft Azure, Oracle Cloud, Computer Vision, Android Studio, Figma, IntelliJ Idea, Cursor, Jenkins